FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026, Computer Vision coursework) with a B.Tech in IT (Neural Networks & Deep Learning) and co-author of a peer-reviewed computer-vision paper on image classification. Built a real-time detector trained end to end and chosen on latency grounds for constrained hardware, and an AI/ML internship where evaluation harnesses and a hand-built ground-truth set drove the fixes. Seeking an Engineer II, Deep Learning role on Shield AI's perception team in Melbourne.

TECHNICAL SKILLS

Computer Vision: OpenCV (image classification, object detection, Haar cascade training, edge detection — Canny/Sobel), IoU evaluation, real-time camera pipelines

Machine Learning: Python (primary), scikit-learn (incl. MLP), XGBoost, pandas, LLM fine-tuning of transformer models (Unsloth, QLoRA, GGUF), Ollama, evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction)

Engineering: Git / GitHub, GitHub Actions CI, Docker, Linux, Colab / Jupyter, Kestra, AWS (CloudFormation, Fargate), FastAPI, SQL

Also: Rust, Kotlin, TypeScript, JavaScript, Bash; Claude Code (skills, hooks, subagents), MCP servers

KEY PROJECTS

Fire Detection & Alerting System · Python, OpenCV, Haar cascades, threading *2023 – 24*
B.Tech capstone, team of 4 — real-time detection from a live camera feed, with off-site alerting

- Trained a custom OpenCV Haar cascade end to end (positive marking, sample vectorisation, 15-stage training, XML export) and chose it over heavier detectors on latency grounds — reaction time beats a few points of accuracy.
- Built the real-time loop: per-frame detection, bounding-box overlay, and threaded alarm and WhatsApp alerting so capture never blocks; targeted sub-5-second response, restart-safe, on a 4 GB / 2.9 GHz minimum spec.

Gemma Delegation System — local/cloud LLM hybrid · Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab *May 2026*

- Fine-tuned a transformer model on curated examples (4-bit QLoRA, LoRA r=16) in Colab and exported GGUF for local inference in Ollama; accept/reject feedback curates the JSONL training set.
- Built the MCP server that delegates lightweight tasks to the local model and fails open to the frontier model.
  ↳ GitHub: github.com/faraaz817/gemma-delegation-system
- **MGM Prayer Alerts — Android app in community use:** wrote functional requirements before coding, reviewed the code against them and fixed a critical AM/PM storage bug; added GitHub Actions to test, build and publish releases. (Kotlin, JUnit) github.com/faraaz817/MgmPrayerAlerts

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers (40 combinations) as a lesion feature-extraction step on a 15,000-image, four-class maize dataset; best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn, ten classifiers). ↳ questjournals.org/jses · Code: github.com/faraaz817/Machine_Learning

EXPERIENCE

AI/ML Engineering Intern — Prodegee *Mar 2026 – Jun 2026*
Remote, Melbourne · RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Trained an XGBoost multi-output tier regressor over 200 villages in 7 Mauritanian Wilayas; documented that its 0.816 "confidence" measured agreement with the training heuristic, not accuracy.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries.

- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and wrote the semester-2 handoff memo.

Tools: Python, scikit-learn, XGBoost, pandas, GeoPandas, Shapely, QGIS, Kestra, Git

Full-Stack Developer Intern — Vayaa *Sep 2026 – present*

Melbourne, VIC · hybrid · company website: Python (FastAPI) + PostgreSQL, React + TypeScript, deployed on AWS

- Built the contact-enquiry API in Rust, then led its port to Python (FastAPI) on the mentor's call, gating the cutover on 50 parity tests passing against both backends.
- Took it live on AWS with CloudFormation (CloudFront → ECS Fargate → RDS PostgreSQL), fixing Free-plan limits, a CloudFront account block and RDS TLS (verify-full); kept a decision log for every design call.

Dispatch Team Member — Amart Furniture *Feb 2025 – Mar 2026*

Melbourne, VIC · part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

EDUCATION

Master of Artificial Intelligence — RMIT University *Jul 2024 – Jul 2026*

Melbourne, VIC · Coursework: **Computer Vision**, Machine Learning, AI Systems Design, Intelligent Decision Making (ASP/Clingo), IoT

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology *Dec 2020 – May 2024*

Hyderabad, India (affiliated to JNTUH) · Coursework incl. Artificial Neural Networks & Deep Learning, Machine Learning, Data Mining

CERTIFICATIONS, AWARDS & LEADERSHIP

- Certificate of Publication, Quest Journals (2023) · Smart India Hackathon delegate, Government of India (2022).
- IBM Z Xplore – Concepts and IBM Z & LinuxONE Community Contributor – Level 1 (IBM digital badges via Credly, 2026).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.